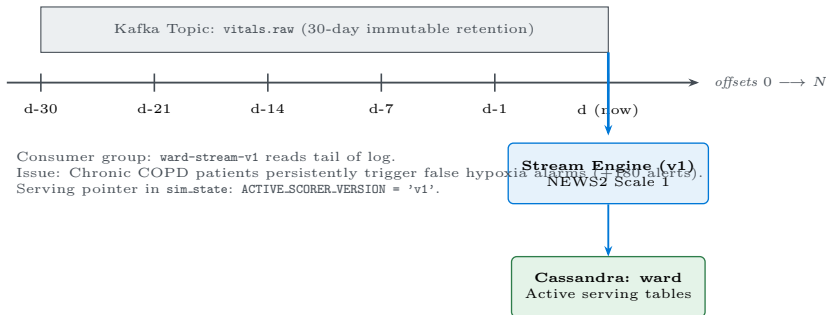
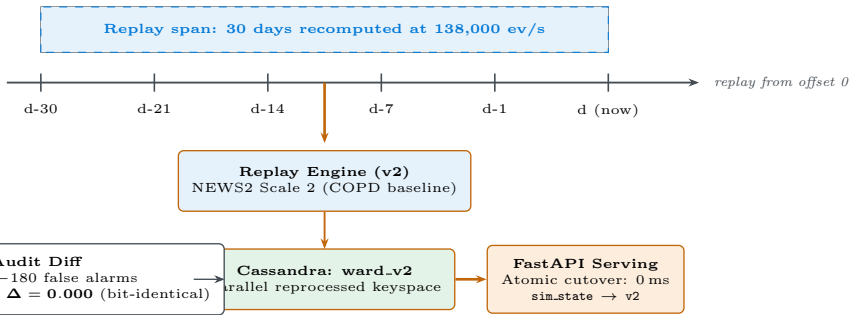


(a) Active Production Stream (v1) — Processing at Log Head ($t = \text{now}$)



(b) Historical Replay (v2) — Reprocessing 30 Days from Offset 0 + Atomic Cutover



Replay consumer group: ward-stream-v2-replay with `auto.offset.reset=earliest`.

Once the clinical audit diff confirms the zero-delta invariant on control patients, an administrator invokes `POST /api/v1/pipeline/cutover`. The serving layer points to ward.v2 instantly with **zero downtime**.